

COMP3278 Group Project — SocialHub

A lightweight social-media web application built around a MySQL relational database. Users can register, post text and images, like/unlike posts, comment, browse profiles, view platform-wide analytics, and query the database in natural language through an AI chat assistant that translates questions into SQL.

1. User Requirements

1.1 Target Users

Two informal personas were considered throughout development:

Persona	Primary needs
End user	Create an account, share text/photo posts, react and comment, discover popular posts.
Platform analyst	Inspect activity trends, identify top posts/users, and run ad-hoc SQL-style questions.

1.2 Functional Requirements

ID	Requirement
F-1	Visitors may register with a unique username, display name, and password.
F-2	Registered users authenticate via username/password; sessions are JWT cookies.
F-3	A user may upload / replace a profile avatar (WebP/JPEG/PNG, ≤ size limit).
F-4	A user may create a post with text (≤ 2000 chars), uploaded image blobs, and/or image URLs.
F-5	Every user can browse the global feed sorted by Latest or Popular (paginated).
F-6	A user can view any other user's profile, showing their posts and aggregate stats.
F-7	A signed-in user may like/unlike a post; each (user, post) pair is unique.
F-8	Any user may read comments on a post; signed-in users may post comments.
F-9	A post author may delete their own post (cascades to images, likes, comments).
F-10	Anyone may access <code>/analytics</code> showing top 10 posts, top 10 users, and 30-day activity.
F-11	A signed-in user may chat with an AI agent that answers questions by running SQL on the DB.

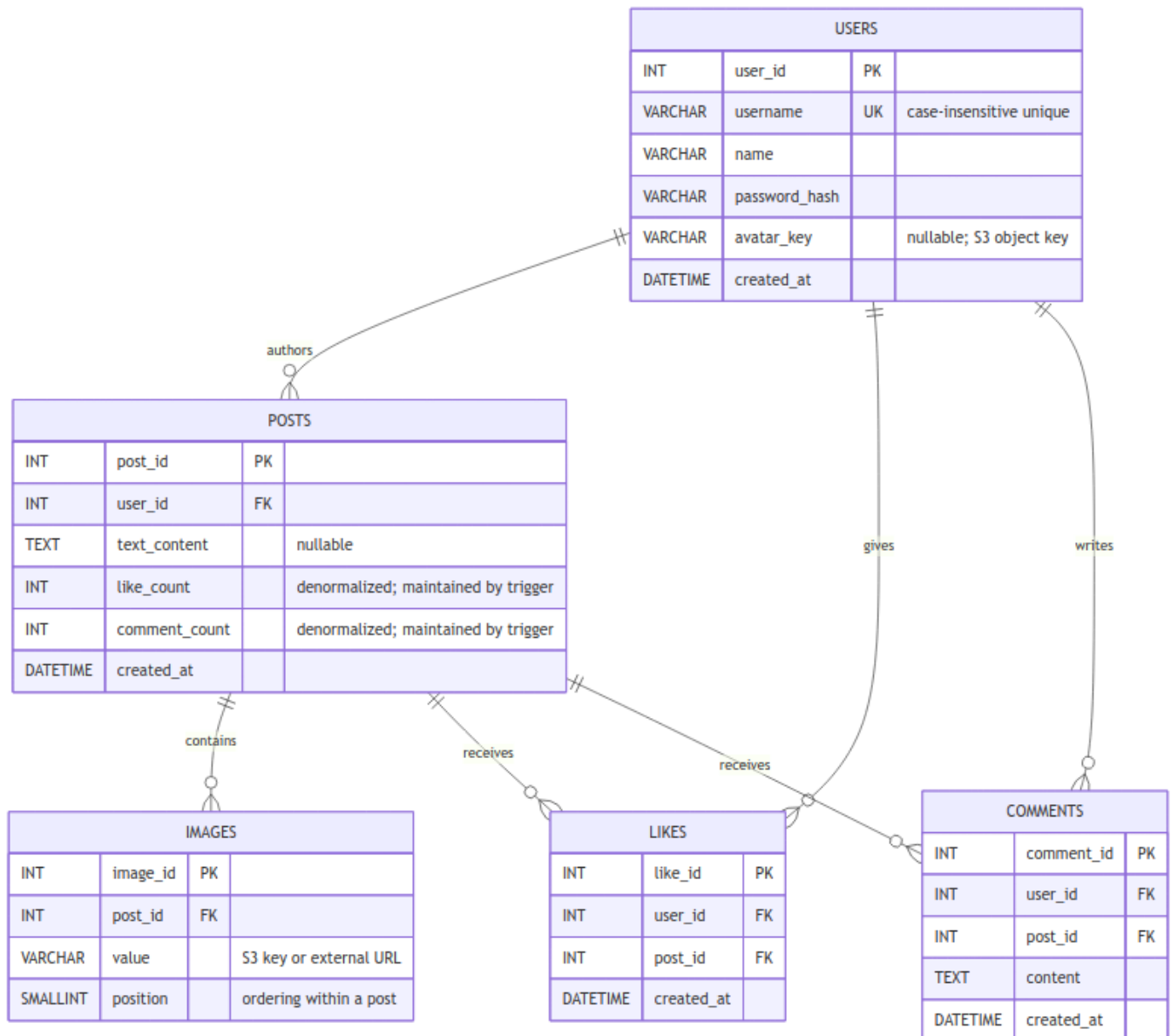
1.3 Non-Functional Requirements

- Consistency** — denormalized counters (`like_count`, `comment_count`) kept in sync by DB triggers, not application code.
- Statelessness** — API is stateless; all session state lives in a signed JWT cookie.
- Scalability** — feeds use keyset (cursor) pagination over indexed columns.

- **Separation of concerns** — React SPA ↔ FastAPI ↔ MySQL/MinIO; containerized, reproducible deployment.
- **Security** — bcrypt password hashing, HttpOnly session cookie, ownership checks on mutating routes, MIME/size validation on uploads.

2. ER Modeling and Design

2.1 Entity-Relationship Diagram



Source: [assets/er-diagram.mmd](#)

2.2 Design Rationale

- **Unique (`user_id`, `post_id`) on `likes`** — enforces idempotent "like" semantics at the DB level.
- **Functional index `LOWER(username)`** — lets users log in case-insensitively while still storing the original casing.
- **Denormalized counters on `posts`** — `like_count` and `comment_count` avoid expensive `COUNT(*)` during feed queries; consistency is guaranteed by four triggers (`after_like_insert/delete`, `after_comment_insert/delete`).

- **ON DELETE CASCADE** — deleting a user or post automatically removes dependent rows; avoids application-level fan-out.
- **Composite keyset indexes** — `(created_at DESC, post_id)` and `(like_count DESC, post_id)` support stable, $O(\log n)$ cursor pagination even when values tie.
- **images.value** — single column storing either an internal `s3://...` reference or an external URL, keeping the schema flat while supporting both upload paths.

2.3 Normalization Notes

All tables are in **3NF**. The only non-key derived data (`like_count`, `comment_count`) is a deliberate denormalization: the cost (four tiny triggers) is tiny compared with the benefit ($O(1)$ reads on the hot feed path).

3. SQL Queries

All SQL below is executed through `aiomysql` in `app/core/queries.py`. Placeholders (`%s`) are parameterized — no string interpolation of user input.

3.1 Schema (excerpt)

```
CREATE TABLE users (
  user_id      INT PRIMARY KEY AUTO_INCREMENT,
  username     VARCHAR(39) NOT NULL,
  name         VARCHAR(50) NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  avatar_key    VARCHAR(255) DEFAULT NULL,
  created_at   DATETIME NOT NULL DEFAULT NOW(),
  UNIQUE KEY users_username_lower ((LOWER(username)))
);

CREATE TABLE posts (
  post_id      INT PRIMARY KEY AUTO_INCREMENT,
  user_id      INT NOT NULL,
  text_content TEXT,
  like_count   INT NOT NULL DEFAULT 0,
  comment_count INT NOT NULL DEFAULT 0,
  created_at   DATETIME NOT NULL DEFAULT NOW(),
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);

CREATE TRIGGER after_like_insert AFTER INSERT ON likes FOR EACH ROW
  UPDATE posts SET like_count = like_count + 1 WHERE post_id = NEW.post_id;
```

(See `schema.sql` for the complete DDL including `images`, `likes`, `comments`, the four triggers and all indexes.)

3.2 Authentication

```
-- Register: uniqueness check uses the LOWER(username) functional index
SELECT * FROM users WHERE LOWER(username) = LOWER(%s);

-- Register: insert
INSERT INTO users (username, name, password_hash) VALUES (%s, %s, %s);
```

3.3 Feed — Latest (keyset pagination)

```
SELECT p.*, u.username, u.name, u.avatar_key
FROM posts p JOIN users u ON p.user_id = u.user_id
WHERE (p.created_at, p.post_id) < (%s, %s)
ORDER BY p.created_at DESC, p.post_id DESC
LIMIT %s;
```

3.4 Feed — Popular

```
SELECT p.*, u.username, u.name, u.avatar_key
FROM posts p JOIN users u ON p.user_id = u.user_id
WHERE (p.like_count, p.post_id) < (%s, %s)
ORDER BY p.like_count DESC, p.post_id DESC
LIMIT %s;
```

3.5 Batch Hydration of a Feed Page

Two batched queries avoid the N+1 problem when rendering a page of posts:

```
-- Images for all posts on the current page
SELECT * FROM images
WHERE post_id IN (%s, %s, ...)
ORDER BY post_id ASC, position ASC;

-- Which of these posts has the current user already liked?
SELECT post_id FROM likes
WHERE user_id = %s AND post_id IN (%s, %s, ...);
```

3.6 Profile Summary (aggregate)

```
SELECT u.user_id, u.username, u.name, u.avatar_key, u.created_at,
       COUNT(DISTINCT p.post_id) AS post_count,
       COALESCE(SUM(p.like_count), 0) AS total_likes
FROM users u LEFT JOIN posts p ON p.user_id = u.user_id
WHERE LOWER(u.username) = LOWER(%s)
GROUP BY u.user_id, u.username, u.name, u.avatar_key, u.created_at;
```

3.7 Analytics Queries

```
-- Top 10 most liked posts
SELECT p.post_id, u.username, p.text_content, p.like_count
FROM posts p JOIN users u ON p.user_id = u.user_id
ORDER BY p.like_count DESC LIMIT 10;

-- Top 10 most active users (+ likes received)
SELECT u.username, u.name,
       COUNT(DISTINCT p.post_id) AS post_count,
       COALESCE(SUM(p.like_count), 0) AS total_likes
FROM users u LEFT JOIN posts p ON p.user_id = u.user_id
GROUP BY u.user_id, u.username, u.name
ORDER BY post_count DESC LIMIT 10;

-- Posts / likes per day (last 30 days)
SELECT DATE(created_at) AS date, COUNT(*) AS count
FROM posts WHERE created_at >= NOW() - INTERVAL 30 DAY
GROUP BY DATE(created_at) ORDER BY date ASC;

SELECT DATE(created_at) AS date, COUNT(*) AS count
FROM likes WHERE created_at >= NOW() - INTERVAL 30 DAY
GROUP BY DATE(created_at) ORDER BY date ASC;
```

3.8 Maintenance — Recompute Counts

If triggers are ever bypassed (e.g., bulk-loading seed data), a single `UPDATE ... JOIN` statement resynchronises `posts`:

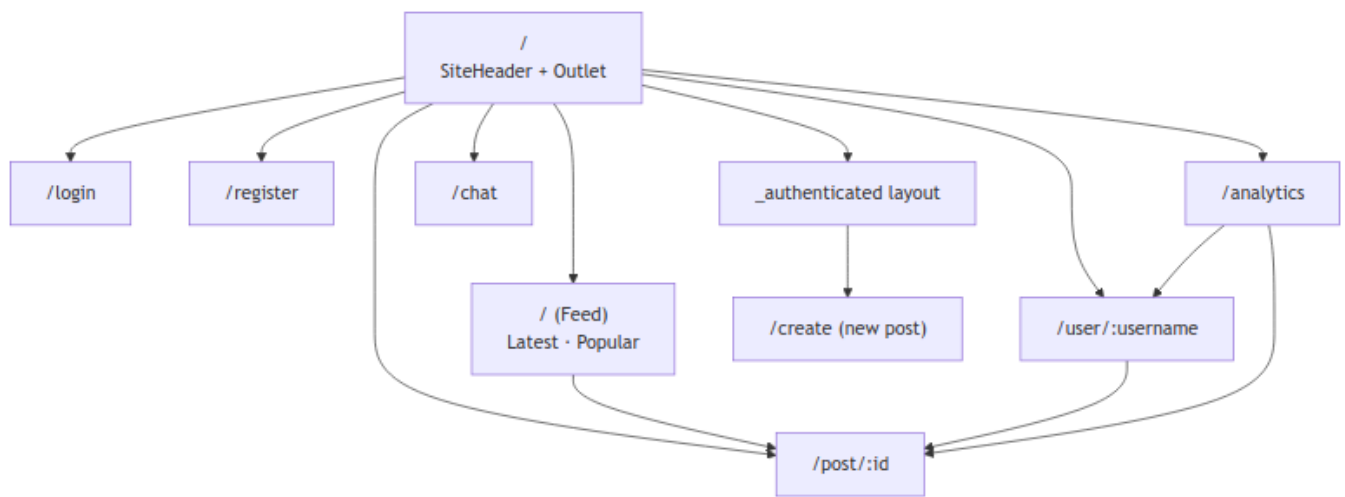
```
UPDATE posts p
LEFT JOIN (SELECT post_id, COUNT(*) AS cnt FROM likes GROUP BY post_id) lk ON
lk.post_id = p.post_id
LEFT JOIN (SELECT post_id, COUNT(*) AS cnt FROM comments GROUP BY post_id) cm ON
cm.post_id = p.post_id
SET p.like_count = COALESCE(lk.cnt, 0),
    p.comment_count = COALESCE(cm.cnt, 0);
```

4. UI Design

4.1 Visual Language

The front-end uses a **retro "8-bit" design system** — shadcn/ui primitives re-skinned with pixel fonts, hard-edged borders, and a chunky palette. Icons come from [pixelarticons](#). Typography is toggleable (pixel ↔ sans) via a header button; dark mode is supported.

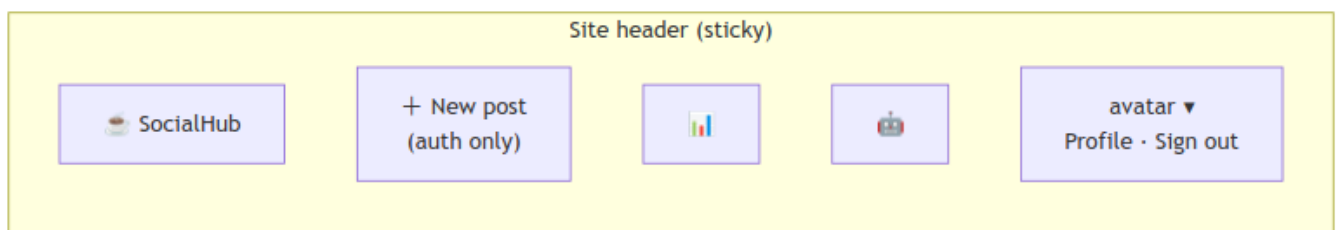
4.2 Information Architecture



Source: *assets/information-architecture.mmd*

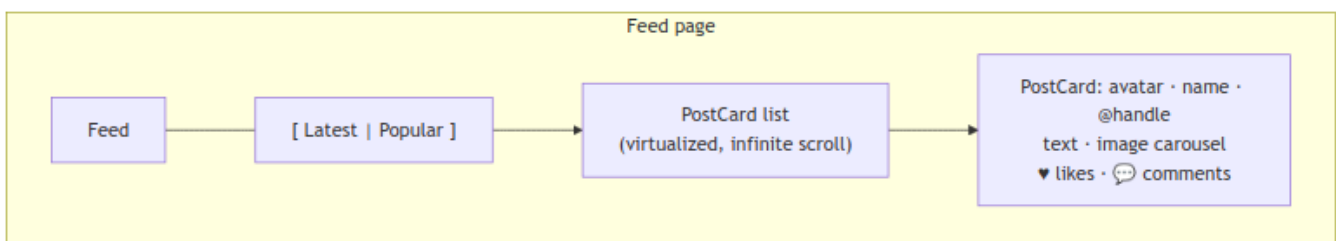
4.3 Page-Level Sketches

Site header



Source: *assets/ui-site-header.mmd*

Feed page



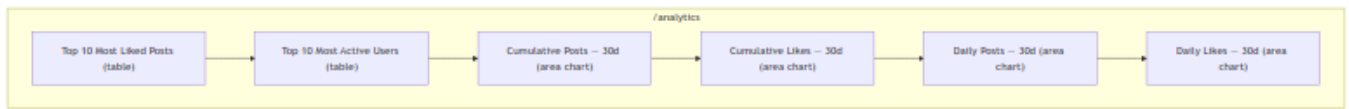
Source: *assets/ui-feed.mmd*

User profile



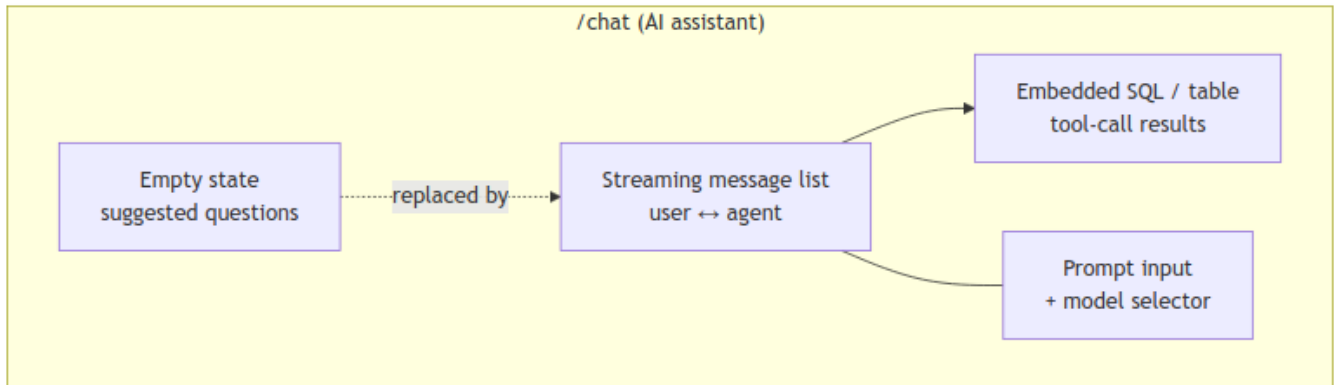
Source: *assets/ui-profile.mmd*

Analytics



Source: *assets/ui-analytics.mmd*

Chat

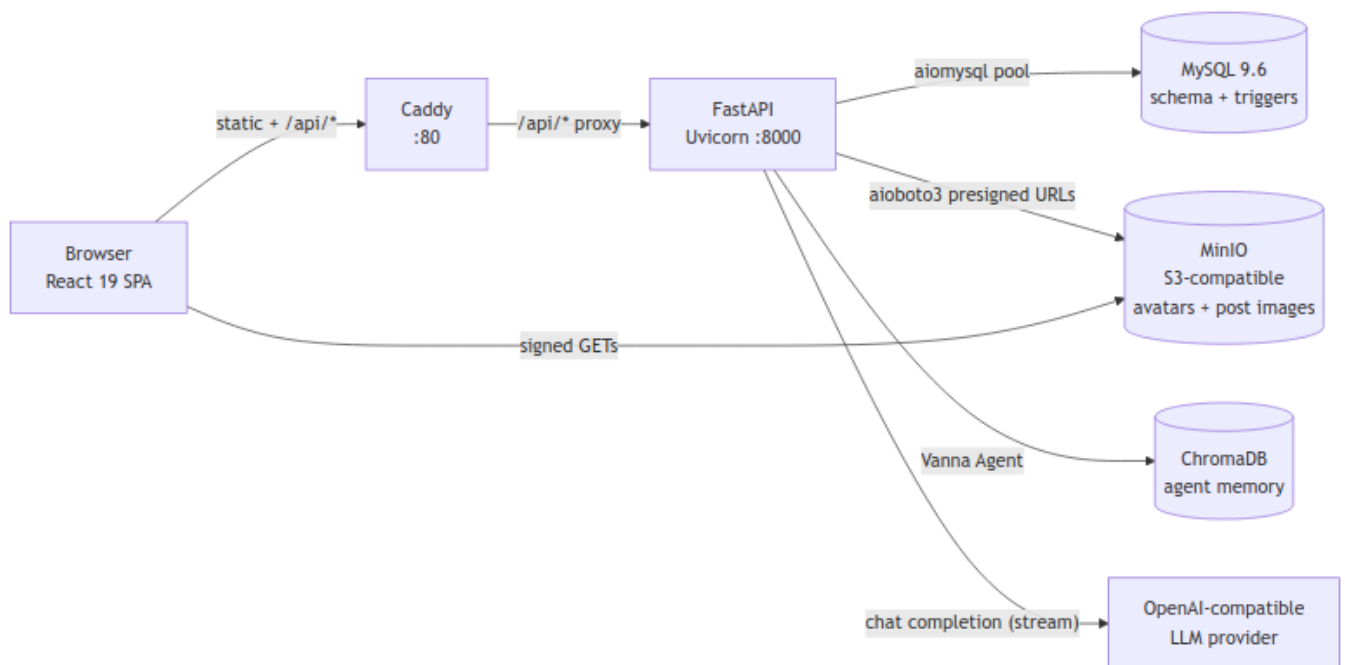


Source: *assets/ui-chat.mmd*

4.4 UX Highlights

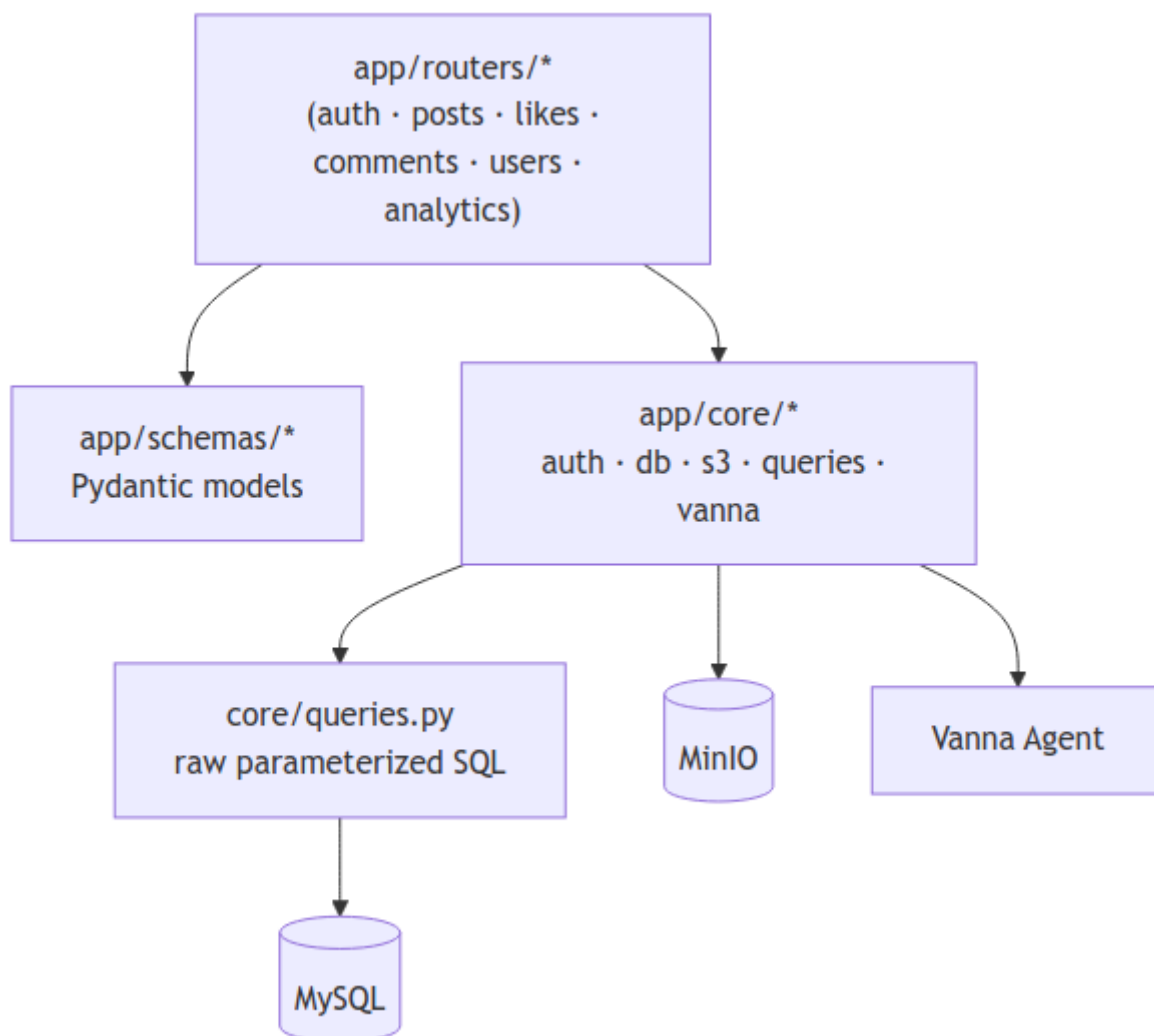
- **Optimistic likes** — the heart toggles immediately; TanStack Query rolls back on failure.
- **Infinite scroll** — cursors returned by the API drive `useInfiniteQuery`; a virtualizer keeps the DOM small.
- **Post composition** — supports drag-and-drop blob uploads *and* external URLs in one form; server validates MIME type and size.
- **Auth-aware header** — shows Sign in / Sign up for guests, and avatar menu for signed-in users.
- **Resilient errors** — root `errorComponent` renders a friendly page and a "back to home" button.
- **Toasts** — `sonner` provides non-blocking success/error feedback for mutations.

5. System Architecture



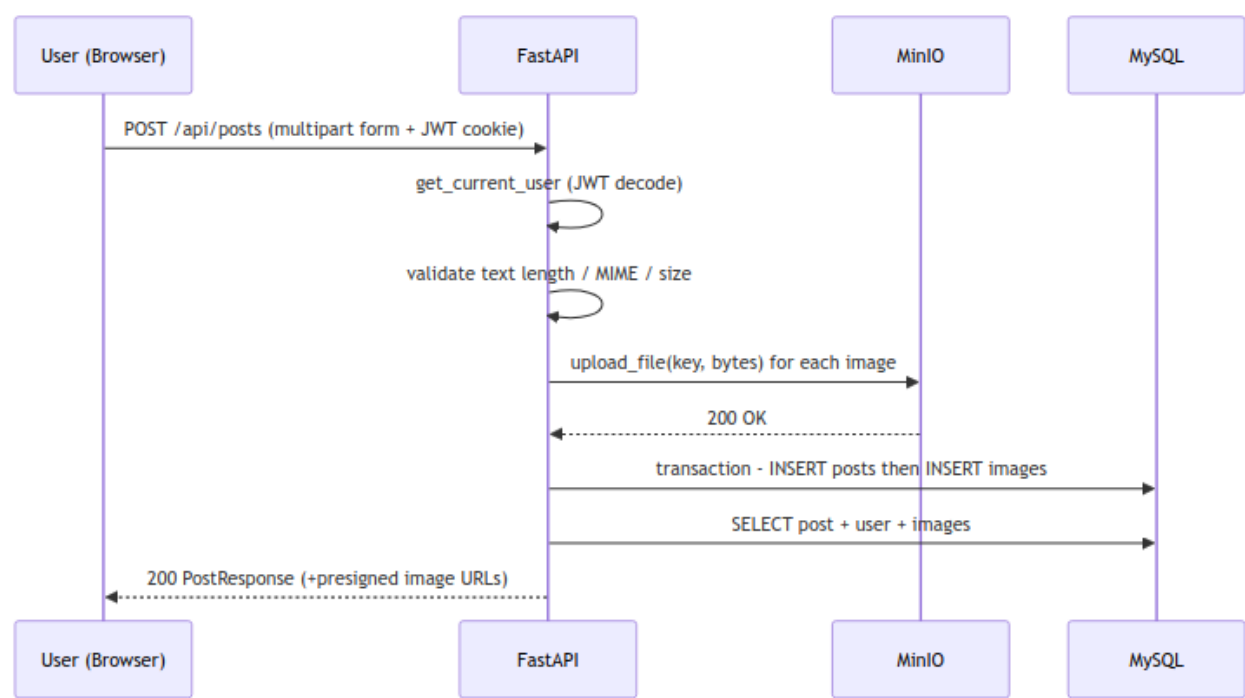
Source: *assets/system-architecture.mmd*

5.1 Backend Layer Map



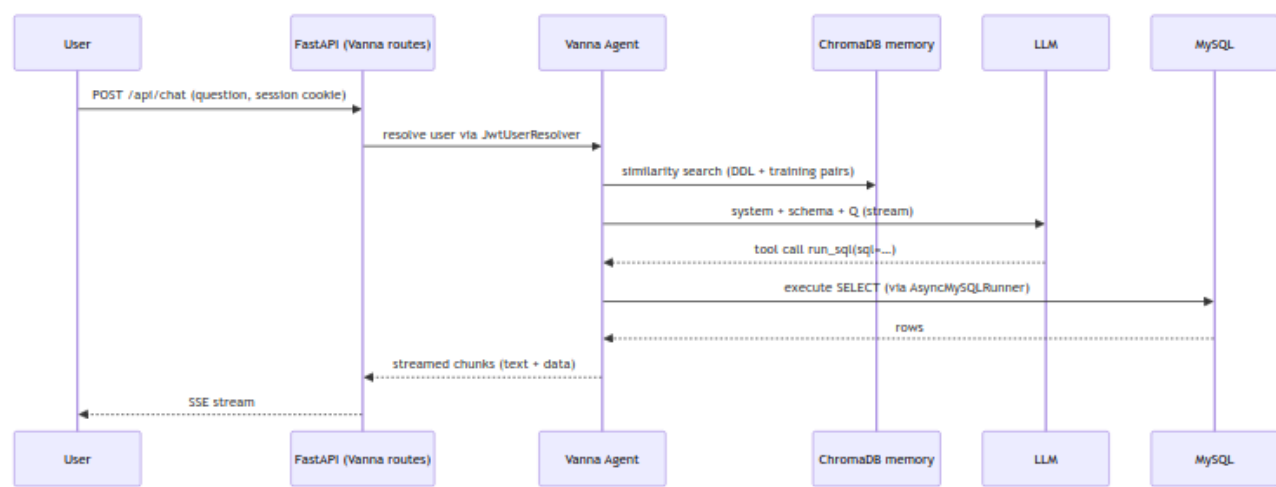
Source: *assets/backend-layers.mmd*

5.2 Request Flow — "Create Post"



Source: *assets/sequence-create-post.mmd*

5.3 AI Chat Flow — "Natural-language → SQL"



Source: *assets/sequence-ai-chat.mmd*

The agent is seeded at startup (`seed_memory()`) with the schema DDL and 16 curated (`question`, `SQL`) training pairs (see `app/core/vanna_config.py`) so that common analytics questions resolve directly from memory.

A full design document for this subsystem — grounding, request lifecycle, safety, per-request model selection, streaming protocol — lives in `TEXT_TO_SQL.md`.

6. Technology Stack

The table below is a map; each library is described in depth in §6.1–§6.5 with the exact role it plays in SocialHub.

Layer	Choice
Frontend core	React 19, TypeScript 5, Vite, Bun
Frontend state	TanStack Router, TanStack Query, TanStack Virtual
Frontend UI	Tailwind CSS v4, shadcn/ui (8-bit theme), Radix UI, pixelarticons, Recharts, Sonner, Streamdown, Motion, Embla Carousel
API client	openapi-fetch + openapi-typescript
Backend core	Python 3.14, FastAPI, Uvicorn, Pydantic v2, structlog
DB driver	aiomysql — raw, parameterized SQL (no ORM)
Object store	MinIO (S3-compatible) accessed via aioboto3 with presigned URLs
Auth	bcrypt (hashing) + PyJWT in an HttpOnly cookie
AI / text-to-SQL	Vanna Agent, ChromaDB, AsyncOpenAI, pandas
Database	MySQL 9.6 (schema + triggers in schema.sql)
Reverse proxy	Caddy (serves SPA + proxies /api/* in the deploy profile)
DevX	uv , Ruff, Bun, Oxlint/Oxfmt, pytest, vitest, Docker Compose

6.1 Backend Libraries

- **FastAPI (≥ 0.135)** — Async web framework built on Starlette. Declarative routing (`@router.post(...)`), dependency injection via `Depends()`, automatic OpenAPI/Swagger generation, and typed request/response models. Used everywhere in `app/routers/*`; each endpoint is an async function that receives Pydantic-parsed bodies and auth principals.
- **Uvicorn (with [\[standard\] extras](#))** — ASGI server that runs the FastAPI app. `--reload` is used in development; in Docker we run it directly under `entrypoint.sh`.
- **Pydantic v2 + [pydantic-settings](#)** — Data-validation layer. All request bodies and response models in `app/schemas/*` are Pydantic models — invalid input is rejected at the edge with a 422 before any business code runs. `pydantic-settings` loads `.env` values into a typed `Settings` object (`app/core/config.py`).
- **aiomysql (≥ 0.3)** — Truly-async MySQL client. We intentionally avoid an ORM: every SQL statement lives in `app/core/queries.py`, parameterized with `%s`. A single `aiomysql.Pool` is initialised in the FastAPI `lifespan` and checked out per request via `db.get_conn()` / `db.transaction()` context managers.
- **aioboto3 (≥ 15.5)** — Async port of `boto3`. Used by `app/core/s3.py` to upload avatars / post images to MinIO and to mint presigned `GET` URLs the browser hits directly — the FastAPI process never streams image bytes in responses.

- **PyJWT (≥ 2.12)** — HS256 signed session tokens. Tokens are set as an `HttpOnly` cookie on login/register, decoded in the `get_current_user` dependency, and also parsed inside the Vanna `JwtUserResolver` to reuse the same identity for the AI chat.
- **bcrypt (≥ 5.0)** — Password hashing with per-hash salt and a tunable work factor. `auth.hash_password` / `auth.verify_password` are the only surfaces that touch plaintext passwords, and plaintext never leaves request memory.
- **python-multipart (≥ 0.0.22)** — Enables FastAPI to parse `multipart/form-data` bodies. Used by the post-create and avatar-upload endpoints to accept files alongside JSON-ish form fields.
- **cryptography (≥ 43)** — A transitive-but-explicit dependency pulled in for MySQL's `caching_sha2_password` auth plugin — without it connecting to MySQL 9 fails.
- **structlog (≥ 25.5)** — Structured, key-value logging. Configured in `main.py` with `contextvars.merge_contextvars`, ISO timestamps, and a colorised console renderer; every route logs events like `log.info("post_deleted", post_id=..., user_id=...)`.
- **Vanna (vanna[openai] ≥ 2.0)** — The agent framework that drives the `/chat` page. Supplies the agent loop, tool registry, OpenAI integration, ChromaDB memory adapter, SSE chat routes, and rich UI component protocol. We subclass a few of its building blocks (`OpenAILlmService`, `RunSqlTool`, `ChatHandler`, `UserResolver`) rather than reinventing them — see `TEXT_TO_SQL.md`.
- **ChromaDB (≥ 1.5)** — Embedded persistent vector store used for the Vanna agent's memory: the schema DDL plus curated question/SQL pairs are upserted as documents and retrieved by similarity at chat time. Persists under `.chroma/`.
- **pandas** — `AsyncMySQLRunner.run_sql` returns a `pandas.DataFrame` because that's what Vanna's `SqlRunner` contract requires; the DataFrame is serialised to `{columns, rows}` before streaming to the browser.
- **openai (AsyncOpenAI client)** — Official OpenAI SDK, configured with a custom `base_url` so any OpenAI-compatible endpoint works. We use the `async` client (`AsyncOpenAI`) exclusively — the stock Vanna integration uses the sync client, which would block Uvicorn's event loop on every chat call.

6.2 Backend Dev / Quality Tools

- **uv** — Fast Rust-based Python package & project manager. `pyproject.toml` declares dependency groups (`dev`); `uv sync --all-groups` installs both runtime and dev dependencies into a project-local `.venv`. `uv run ...` is used instead of `python -m`.
- **Ruff (≥ 0.15)** — All-in-one Python linter/formatter written in Rust. `ruff.toml` picks rule families `E/F/I/UP/B/SIM/ASYNC` (errors, imports, modernization, bugbear, simplifications, async pitfalls) with a 100-char line limit and Python 3.14 target.
- **pytest + pytest-asyncio + httpx + faker + tqdm** — Backend test stack. `asyncio_mode = "auto"` lets every coroutine test run without per-function decorators; `httpx` powers FastAPI's `TestClient` equivalent; `faker` generates the demo dataset in `scripts/seed.py`.

6.3 Frontend Libraries

- **React 19** — UI library. Used with the new Suspense-friendly data-fetching patterns; the whole app is an SPA mounted in `main.tsx`.
- **TanStack Router (≥ 1.168)** — Typed, file-based router (`src/routes/*.tsx`). The generated `routeTree.gen.ts` gives compile-time URL / param safety. We use a nested `_authenticated` layout route as an auth gate, `$id / $username` dynamic segments, and a `validateSearch` schema for `/chat?q=...` deep-links.

- **TanStack Query (≥ 5.96)** — Server-state cache for all `/api/*` reads and mutations. We use `useInfiniteQuery` for feed / comment pagination, `useMutation` with optimistic updates for likes, and `invalidateQueries` after mutations to keep feeds fresh.
- **TanStack Virtual** — Row virtualization for the feed: only the posts currently in the viewport are kept in the DOM, so scrolling large feeds stays smooth.
- **openapi-fetch + openapi-typescript** — Our typed API client. `bun run generate` hits the running backend's `/openapi.json` and regenerates `src/lib/api/schema.d.ts`. Code calls e.g. `client.GET("/api/posts", { params: { query: { sort } } })` and the types for every endpoint (params, body, responses) come from the server, eliminating drift.
- **Tailwind CSS v4** — Utility-first CSS; configured through the `@tailwindcss/vite` plugin (no `tailwind.config.js` — v4 uses a CSS-first config in `index.css`).
- **shadcn/ui (8-bit theme)** — A set of component *source files* copied into `src/components/ui` rather than a classic `node_modules` library. We use the retro "8-bit" preset, which restyles Radix primitives with pixel fonts and hard borders.
- **Radix UI** — Unstyled, accessible primitives (dropdown, tabs, dialog, avatar, separator, ...) that shadcn wraps. Provides keyboard navigation and focus management out of the box.
- **pixelarticons** — Single-stroke pixel-art icon set; imported as React components (`import { Heart, Fire } from "pixelarticons/react"`). Fits the 8-bit visual theme.
- **Recharts (3.8)** — Composable chart primitives (`AreaChart`, `CartesianGrid`, ...). Powers every chart on `/analytics`; wrapped in shadcn's `ChartContainer` so styles flow from CSS variables.
- **Sonner** — Minimal toast library. Mounted once in `__root.tsx`; mutations call `toast.success(...)` / `toast.error(...)` for non-blocking feedback.
- **Streamdown** — Markdown renderer tuned for *streaming* tokens — can render partial / unterminated Markdown safely. Used to render the assistant's streamed replies in `/chat` without flicker.
- **Motion (framer-motion successor)** — Animations, used sparingly for micro-interactions (e.g. the chat scroll-to-bottom pill).
- **Embla Carousel** — Lightweight carousel used inside `PostCard` when a post has more than one image.
- **date-fns (v4)** — Functional date library; `formatDistanceToNow(parseApiDate(...))` is how all "2 minutes ago" labels are rendered.
- **nanoid** — Short unique id generator; used for client-side ids on chat messages.
- **ai (Vercel AI SDK)** — Types and helpers for chat UIs; we reuse its message-shape primitives rather than invent our own.
- **class-variance-authority + tailwind-merge + clsx** — Conventional shadcn utilities for composing variant-aware class strings without conflicts.

6.4 Frontend Dev / Quality Tools

- **Bun** — JavaScript runtime + package manager + test runner. Replaces npm/yarn/pnpm; `bun install`, `bun dev`, `bun test`, `bun run generate` are the daily commands.
- **Vite 8** — Dev server (HMR in ~10 ms) and production bundler. Proxies `/api` to the FastAPI backend during development via `vite.config.ts`.
- **Vitest + @testing-library/react + happy-dom** — Unit test stack for React components; `happy-dom` gives a faster, smaller DOM than jsdom.
- **MSW** — Mocks network requests in component tests so we exercise loading / error states without a live API.
- **oxlint / oxfmt** — Rust-based alternatives to ESLint/Prettier that are an order-of-magnitude faster. Configured as `bun lint` / `bun fmt`.

- **knip** — Dead-code / unused-export detector; configured via `knip.json`.
- **openapi-typescript (≥ 7.13)** — Generator that produces `schema.d.ts` from the backend's OpenAPI document (invoked by `bun run generate`).

6.5 Infrastructure & Tooling

- **MySQL 9.6** — Primary datastore. `schema.sql` is mounted into `/docker-entrypoint-initdb.d/` so a fresh container applies the schema + triggers on first boot. Functional index `LOWER(username)` requires MySQL 8+.
- **MinIO** — S3-compatible object store. In local development it replaces AWS S3 byte-for-byte; in production the same `aioboto3` code can point at real S3.
- **Caddy** — Reverse proxy used in the deploy profile. `Caddyfile` serves the compiled SPA statically and proxies `/api/*` to the `api` service. Automatic SPA fallback to `index.html` for client-side routes.
- **Docker Compose** — Orchestrates MySQL + MinIO for local dev, plus the `api` and `web` services under the `deploy` profile for a one-command full-stack deployment.

7. Repository Layout

```
comp3278-social-hub/
├── schema.sql                # DDL + triggers
├── docker-compose.yml        # MySQL, MinIO, API, Caddy (web)
├── Caddyfile                 # SPA fallback + /api proxy
├── pyproject.toml            # Python deps (uv)
├── app/
│   ├── main.py               # FastAPI app + lifespan
│   └── routers/               # auth, posts, likes, comments, users, analytics,
website_config
│   ├── schemas/              # Pydantic request/response models
│   ├── core/
│   │   ├── db.py             # aiomysql pool + transaction ctx
│   │   ├── queries.py         # every SQL statement
│   │   ├── s3.py              # MinIO upload / presign
│   │   ├── auth.py           # bcrypt + JWT
│   │   ├── vanna.py           # AI agent init + memory seeding
│   │   └── vanna_config.py    # DDL + Q/A training pairs
│   └── exceptions.py
├── scripts/seed.py           # Faker-based demo data
├── web/                       # React SPA (Bun)
│   └── src/{routes,components,features/chat,lib}
├── docker/{Dockerfile.api, Dockerfile.web}
└── tests/
```

8. Running the Project

```
# 1. Infrastructure
docker compose up -d          # MySQL + MinIO (schema auto-applied)
```

```
# 2. Backend
uv sync --all-groups
uv run python scripts/seed.py      # optional: Faker-generated demo data
uv run uvicorn app.main:app --reload --host 0.0.0.0 --port 8000

# 3. Frontend
cd web && bun install && bun dev    # http://localhost:5173

# 4. Full containerized deploy
docker compose --profile deploy up --build -d  # http://localhost
```

Quality gates:

```
uv run pytest                # backend tests
uv run ruff check .          # backend lint
cd web && bun test            # frontend tests (vitest)
cd web && bun lint && bun fmt # frontend lint/format
```

9. Summary of Contributions (Database Focus)

1. **Schema design** — 5-table 3NF schema with deliberate denormalization on `posts` for read performance.
2. **Integrity** — foreign keys with `ON DELETE CASCADE`, unique constraints (`users.username`, `likes(user_id, post_id)`), and four triggers keeping counters consistent.
3. **Indexing strategy** — composite indexes tailored to keyset-paginated feed sort orders.
4. **Query design** — all application SQL is centralized in one module, parameterized, and batch-friendly (images and "liked by me" hydrated per page, not per row).
5. **Analytics** — GROUP BY / date-bucket queries power the `/analytics` dashboard.
6. **AI over SQL** — Vanna agent grounded on the same DDL lets users ask the database questions in plain English, with results streamed back and rendered as tables/charts in the chat UI.